

Competition: Chess Move Tracking

Introduction to Digital Image

Chess Move Tracking

Motivation & Requirement

- Design program to detect chess piece
- Use image processing to solve this problem.
- Output Portable Game Notation (PGN) format.
- Can detect chess pieces moving each turn and another side (white-black).
- The algorithm can detect chess on video.
- Visualize its. (Optional)



Fig1. Chess board game



Fig2. Visualize chessboard

Visualize Chess Board Game

Chess piece notation

Chess piece notation



Fig1 King

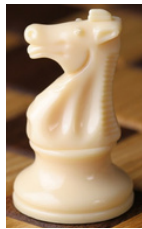


Fig2 Knight



Fig3 Queen

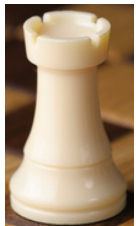


Fig4 Rook



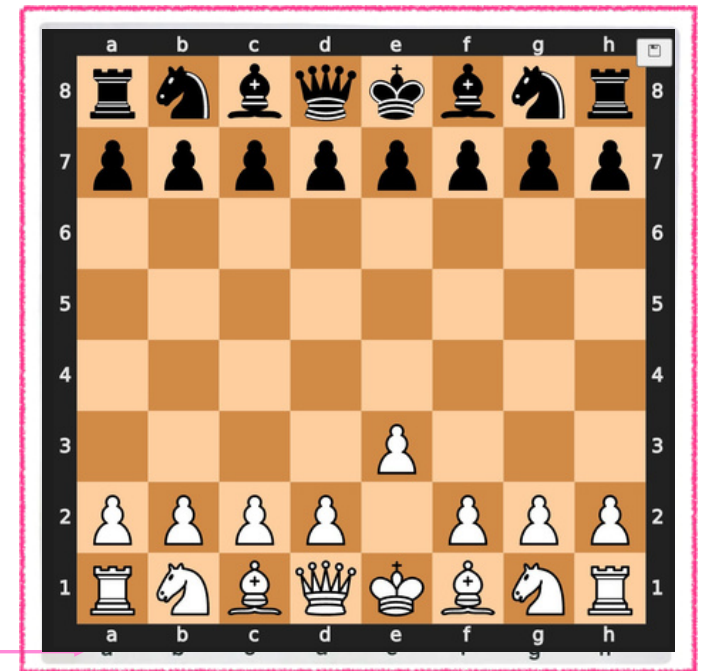
Fig5 Bishop



Fig6 Pawn

Chess piece direction

Chess direction Index



Dataset

Label - PGN

What PGN

- Standard format for recording a game in a text file
- PGN records the sequence

Why PGN

- Most chess programs support it.
- The PGN is a tool that allows players to **replicate games easily** using chess software

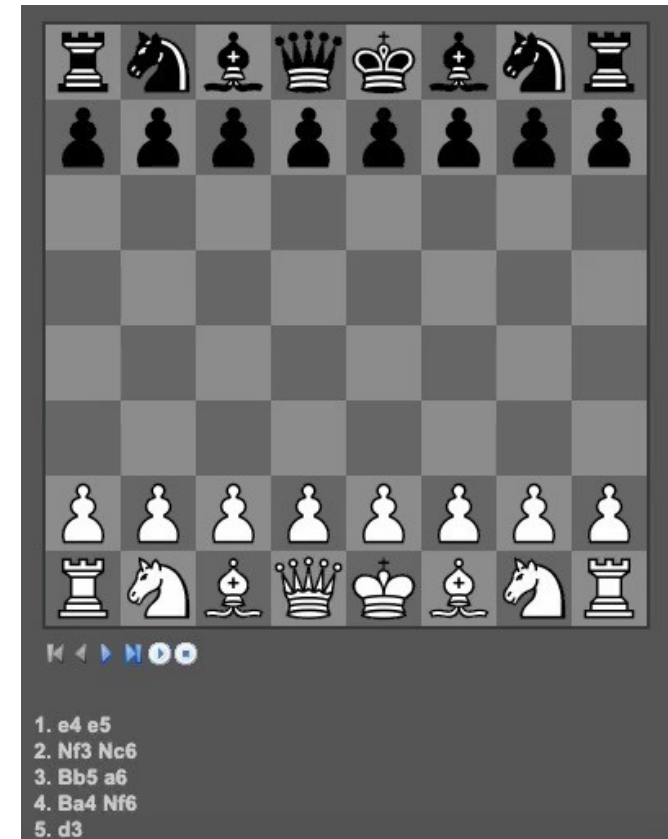
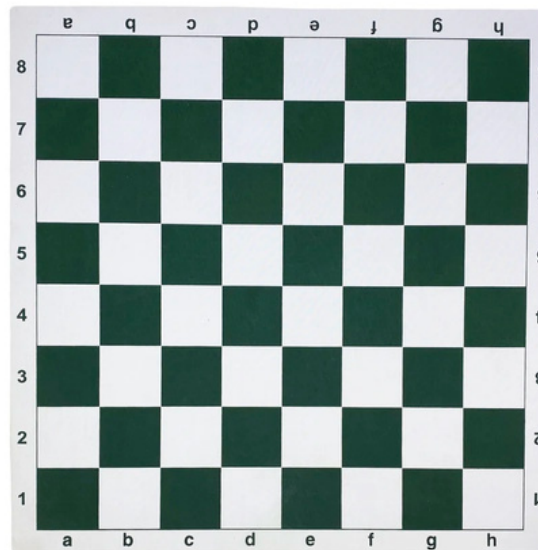


Fig 1. Chess game record history PGN format

Dataset

Label - PGN

Label Video

- PGN Format

§ White always moves first, followed by Black

§ The letter abbreviations are K (king), Q (queen), R (rook), B (bishop), and N (knight). The pawn is given an empty abbreviation

§ x – capture

§ See more :

https://en.wikipedia.org/wiki/Portable_Game_Notation

Start scoring
here

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[Event "Wch U14"]
[Site "Halkidiki GRE"]
[Date "2003.10.23"]
[EventDate "2003.10.23"]
[Round "1.3"]
[Result "0-1"]
[White "Jon Ludvig Hammer"]
[Black "Magnus Carlsen"]
[ECO "A46"]
[WhiteElo "2074"]
[BlackElo "2450"]
[PlyCount "34"]

1. Nf3 d6 2. d4 Nf6 3. Nbd2 g6 4. e4 Bg7 5. Bd3 O-O 6. O-O Nc6
7. c3 e5 8. h3 Nh5 9. dxe5 Nf4 10. Bb5 Nxe5 11. Nxe5 Qg5
12. Ng4 Qxb5 13. Nb3 Ne2+ 14. Kh1 Bxg4 15. hxg4 Rae8 16. Be3
Rxe4 17. Re1 Qh5+ 0-1
```

Fig 1. Example PGN Format

Dataset

Our Videos

Evaluation on Video

- Give 2 moves (Rotation)
- Give 2 moves (Original)
- Give 4 moves
- Give 6 moves + noise
- Give 8 moves



Fig 1. Example Video for 4 Moves (Rotation)



Fig 2. Example Video for 6 Moves

Link to our validation data : <https://www.kaggle.com/t/534af81c1ccd40418cbc11e67db78929>

Evaluation on Kaggle

The Kaggle logo, featuring the word "kaggle" in a light blue, lowercase, sans-serif font with a trademark symbol (TM) to the upper right of the "e". The logo is centered on a dark gray rectangular background.

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